

```
In [26]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("Titanic-Dataset.csv")
print("Dataset Imported")
```

Dataset Imported

```
In [27]: #Explore basic info
print("First Five rows:\n")
print(df.head())
print("\n")
print("Dataset shape: \n")
print(df.shape)
print("\n")
print("Column Info: \n");
print(df.info())
print("\n")
print("Missing Values: \n")
print(df.isnull().sum())
print("\n")
print("Datatypes: \n")
print(df.dtypes);
```

First Five rows:

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	SibSp	\	Name	Sex	Age
0			Braund, Mr. Owen Harris	male	22.0
1					
1			Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0
1					
2			Heikkinen, Miss. Laina	female	26.0
0					
3			Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0
1					
4			Allen, Mr. William Henry	male	35.0
0					

	Parch		Ticket	Fare	Cabin	Embarked
0	0		A/5 21171	7.2500	NaN	S
1	0		PC 17599	71.2833	C85	C
2	0	STON/O2.	3101282	7.9250	NaN	S
3	0		113803	53.1000	C123	S
4	0		373450	8.0500	NaN	S

Dataset shape:

(891, 12)

Column Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

None

Missing Values:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Datatypes:

PassengerId	int64
Survived	int64
Pclass	int64
Name	object
Sex	object
Age	float64
SibSp	int64
Parch	int64
Ticket	object
Fare	float64
Cabin	object
Embarked	object

dtype: object

```
In [28]: #Categorical and Numeric Columns
numeric_cols = df.select_dtypes(include=['number']).columns.tolist()
categorical_cols = df.select_dtypes(include=['object']).columns.tolist()
print("Numeric Columns: \n",numeric_cols)
print("Categorical Columns: \n",categorical_cols)
```

```
Numeric Columns:
['PassengerId', 'Survived', 'Pclass', 'Age', 'SibSp', 'Parch',
'Fare']
Categorical Columns:
['Name', 'Sex', 'Ticket', 'Cabin', 'Embarked']
```

```
In [29]: #Handle Missing Values
#Numeric data -> Median & categorical -> Mode
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
df = df.drop("Cabin",axis=1)
print("Missing values:\n")
print(df.isnull().sum())
```

```
Missing values:
```

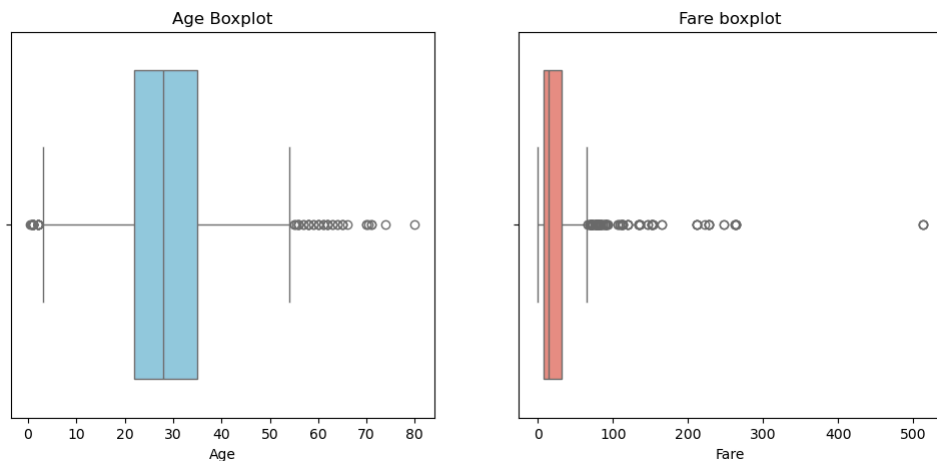
```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
dtype: int64
```

```
In [30]: #Covert Categorical features into numerical features using encoding
#LabelEncoding->Sex column as it consists of only binary Male and Female
#OneHotEncoder -> Embarked as it contains three different class s,c,q
from sklearn.preprocessing import LabelEncoder, OneHotEncoder
le = LabelEncoder()
df['Sex'] = le.fit_transform(df['Sex'])
df = pd.get_dummies(df,columns=["Embarked"],drop_first=True,dtype=int)
print("DONE")
```

```
DONE
```

```
In [31]: #Visualize outliers and remove them
#InterQuartile range Method
import seaborn as sns
fig, axes = plt.subplots(1,2,figsize=(12,5))
sns.boxplot(ax=axes[0],x=df['Age'],color="skyblue")
axes[0].set_title("Age Boxplot")

sns.boxplot(ax=axes[1],x=df['Fare'],color='salmon')
axes[1].set_title("Fare boxplot")
plt.show()
```



```
In [36]: #Remove outliers
outlier_cols = ["Age","Fare"]
for col in outlier_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3-Q1
    lower = Q1-1.5*IQR
    upper = Q3+1.5*IQR
    df = df[(df[col]>=lower) & (df[col]<=upper)]
print(f"Dataset after removing outliers: {df.shape}")
```

Dataset after removing outliers: (705, 12)

In [39]:

```
#Normalize numerical features
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
nums_cols = ['Age', 'Fare']
df[nums_cols] = scaler.fit_transform(df[nums_cols])
print(df.head())
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
2	3	1	3	
3	4	1	1	
4	5	0	3	
5	6	0	3	

		Name	Sex	Age
SibSp	Parch	\		
0		Braund, Mr. Owen Harris	1	-0.636343
1	0			
2		Heikkinen, Miss. Laina	0	-0.214706
0	0			
3		Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	0.733977
1	0			
4		Allen, Mr. William Henry	1	0.733977
0	0			
5		Moran, Mr. James	1	-0.003887
0	0			

	Ticket	Fare	Embarked_Q	Embarked_S
0	A/5 21171	-0.740778	0	1
2	STON/O2. 3101282	-0.689142	0	1
3	113803	2.766613	0	1
4	373450	-0.679580	0	1
5	330877	-0.648346	1	0

In []: